

Report

Collision Avoidance for one moving body with many stationary obstacles

1. How Collision Avoidance Is Implemented in the Code

The model simulates a single moving agent navigating toward a goal while avoiding stationary obstacles. Collision avoidance is achieved through the following steps:

1. Preferred Velocity Calculation

The agent computes a preferred velocity directed toward the goal, limited by its maximum speed.

2. Collision Prediction

For every obstacle, the algorithm calculates:

- relative position
- relative velocity
- combined radius

Using these, it predicts the **time to collision** by solving a quadratic equation.

3. Triggering Avoidance

If a collision is predicted within the agent's time horizon (τ), avoidance is activated.

The algorithm computes a correction vector $\mathbf{u}$ that shifts the agent's velocity out of the Velocity Obstacle (VO).

4. Velocity Adjustment

The agent updates its velocity by subtracting the avoidance vector from the preferred velocity:

This ensures the agent stays outside the collision cone.

5. Safe Trajectory Generation

By repeatedly applying corrected velocities and updating position, the agent follows a smooth, collision-free path around static obstacles.

Modification in original ORCA algorithm

Original ORCA (Multi-Agent)

- Designed for **N-body reciprocal collision avoidance**.
- Each agent shares responsibility, adjusting its velocity by **half** of the required correction:
- Uses **Reciprocal Velocity Obstacles (RVO)** to avoid oscillations in multi-agent motion.
Requires predicting movement of other agents.

Modified Method Used Here (Single-Agent)

- All obstacles are **stationary**, so they cannot share responsibility.
 - The moving agent takes **100% responsibility** for avoiding collisions:
 $v' = v - uv' = v - uv' = v - u$
 - Uses **one-sided Velocity Obstacles (VO)**, not RVO.
 - No reciprocal behavior or interaction with other agents.
 - Simpler, faster, and ideal for environments with static objects.
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Summary

The implemented method computes collision predictions and adjusts the agent's velocity so it remains outside danger regions defined by VO cones. Unlike standard ORCA, which distributes responsibility among multiple moving agents, this approach assigns full responsibility to the single moving agent, making it efficient and effective for navigating around static obstacles.